

FUNCTIONS

Subtitle

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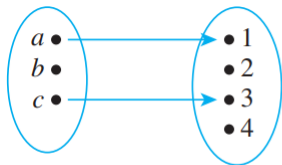
OUTLINE

1. INTRODUCTION TO FUNCTIONS
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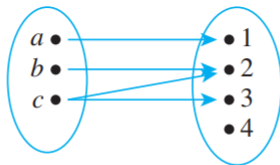
Function

Definition 1

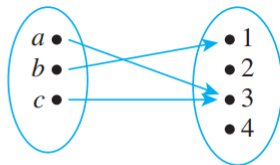
Let A and B be nonempty sets. A function f from A to B is an assignment of exactly one element of B to each element of A . We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A . If f is a function from A to B , we write $f : A \rightarrow B$.



(a)



(b)

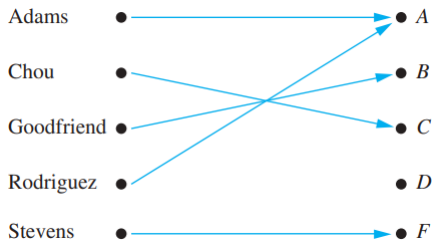


(c)

Only (c) defines a function from $X = \{a, b, c\}$ to $Y = \{1, 2, 3, 4\}$. In (a) the element b in X is not related to any element of Y because there is no arrow that points from b to an element in Y . And in (b) the element c is not related to a unique element of Y because from c there are two arrows that point to two different elements of Y - one toward 2 and the other toward 3.

Definition 2

If f is a function from A to B , we say that A is the **domain** of f and B is the **co-domain** of f . If $f(a) = b$, we say that b is the image of a and a is a **pre-image** of b . The **range**, or **image**, of f is the set of all images of elements of A . Also, if f is a function from A to B , we say that f **maps** A to B .



Example

Let R be the relation with ordered pairs $(Abdul, 22)$, $(Brenda, 24)$, $(Carla, 21)$, $(Desire, 22)$, $(Eddie, 24)$, and $(Felicia, 22)$. Here each pair consists of a graduate student and this student's age. Specify a function determined by this relation.

Solution

If f is a function specified by R , then $f(Abdul) = 22$, $f(Brenda) = 24$, $f(Carla) = 21$, $f(Desire) = 22$, $f(Eddie) = 24$, and $f(Felicia) = 22$. [Here, $f(x)$ is the age of x , where x is a student.] For the domain, we take the set Abdul, Brenda, Carla, Desire, Eddie, Felicia. We also need to specify a codomain, which needs to contain all possible ages of students. Because it is highly likely that all students are less than 100 years old, we can take the set of positive integers less than 100 as the codomain.

Example

Let f be the function that assigns the last two bits of a bit string of length 2 or greater to that string. For example, $f(11010) = 10$. Then, the domain of f is the set of all bit strings of length 2 or greater, and both the codomain and range are the set $00, 01, 10, 11$.

Example

Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$ assign the square of an integer to this integer. Then, $f(x) = x^2$, where the domain of f is the set of all integers, the codomain of f is the set of all integers, and the range of f is the set of all integers that are perfect squares, namely, $0, 1, 4, 9, \dots$

Let f_1 and f_2 be functions from A to $\mathbb{R}$. Then $f_1 + f_2$ and $f_1 f_2$ are also functions from A to $\mathbb{R}$ defined for all $x \in A$ by

$$(f_1 + f_2)(x) = f_1(x) + f_2(x)$$

,

$$(f_1 f_2)(x) = f_1(x) f_2(x)$$

.

Note that the functions $f_1 + f_2$ and $f_1 f_2$ have been defined by specifying their values at x in terms of the values of f_1 and f_2 at x .

Example

Let f_1 and f_2 be functions from $\mathbb{R}$ to $\mathbb{R}$ such that $f_1(x) = x^2$ and $f_2(x) = x - x^2$. What are the functions $f_1 + f_2$ and $f_1 f_2$?

Solution

From the definition of the sum and product of functions, it follows that

$$(f_1 + f_2)(x) = f_1(x) + f_2(x) = x^2 + (x - x^2) = x$$

and

$$(f_1 f_2)(x) = x^2(x - x^2) = x^3 - x^4$$

When f is a function from A to B , the image of a subset of A can also be defined.

Let f be a function from A to B and let S be a subset of A . The image of S under the function f is the subset of B that consists of the images of the elements of S . We denote the image of S by $f(S)$, so $f(S) = \{t \mid \exists s \in S (t = f(s))\}$. We also use the shorthand $\{f(s) \mid s \in S\}$ to denote this set.

The notation $f(S)$ for the image of the set S under the function f is potentially ambiguous. Here, $f(S)$ denotes a set, and not the value of the function f for the set S .

Example

Let $A = \{a, b, c, d, e\}$ and $B = \{1, 2, 3, 4\}$ with $f(a) = 2$, $f(b) = 1$, $f(c) = 4$, $f(d) = 1$, and $f(e) = 1$. The image of the subset $S = \{b, c, d\}$ is the set $f(S) = \{1, 4\}$.

Definition

One-to-One Function

A function f is said to be **one-to-one**, or an **injection**, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f . A function is said to be **injective** if it is one-to-one.

NB: A function f is one-to-one if and only if $f(a) \neq f(b)$ whenever $a \neq b$. This way of expressing that f is one-to-one is obtained by taking the contrapositive of the implication in the definition.

Using quantifiers

$$\forall a \forall b (f(a) = f(b) \rightarrow a = b)$$

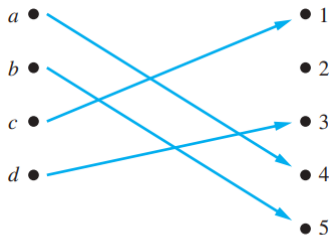
, or equivalently

$$\forall a \forall b (a \neq b \rightarrow f(a) \neq f(b))$$

Example

Determine whether the function f from a, b, c, d to $1, 2, 3, 4, 5$ with $f(a) = 4$, $f(b) = 5$, $f(c) = 1$, and $f(d) = 3$ is one-to-one.

Solution The function f is one-to-one because f takes on different values at the four elements of its domain.



A one-to-one function.

Example

Determine whether the function $f(x) = x^2$ from the set of integers to the set of integers is one-to-one.

Solution: The function $f(x) = x^2$ is not one-to-one because, for instance, $f(1) = f(-1) = 1$, but $1 \neq -1$.

Example

Determine whether the function $f(x) = x^2$ from $\mathbb{Z}^+ \times \mathbb{Z}^+$ is one-to-one.

Solution: The function $f(x) = x^2$ from $\mathbb{Z}^+ \times \mathbb{Z}^+$ is one-to-one, $\forall x \forall y (f(x) = f(y) \rightarrow x = y)$.

Example

Show that the function $f : \mathbb{R} - 0 \rightarrow \mathbb{R}$ defined as $f(x) = \frac{1}{x} + 1$ is injective.

Proof:

Suppose that $f : \mathbb{R} - \{0\}$ and $f(a) = f(b)$.

This means

$$\begin{aligned}\frac{1}{a} + 1 &= \frac{1}{b} + 1 \\ \frac{1}{a} &= \frac{1}{b} \\ a &= b\end{aligned}$$

The function $f(x) = \frac{1}{x} + 1$ is one-to-one.

Example

Prove that $f : [-2, \infty) \rightarrow \mathbb{R}$ given by $f(x) = \sqrt{x+2}$ is injective.

Proof:

Suppose that $f(a) = f(b)$ for some $a, b \in [-2, \infty)$.

Then, $\sqrt{a+2} = \sqrt{b+2}$

$$\sqrt{a+2} = \sqrt{b+2}$$

$$a+2 = b+2$$

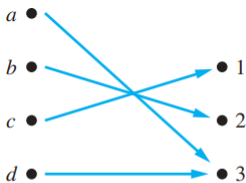
$$a = b$$

The function $f(x) = \sqrt{x+2}$ is one-to-one.

Onto

A function f from A to B is called **onto**, or a **surjection**, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$. A function f is called surjective if it is onto.

A function f is onto if $\forall y \exists x (f(x) = y)$, where the domain for x is the domain of the function and the domain for y is the codomain of the function.



An onto function.

Example Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3\}$ defined by $f(a) = 3$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$. Is f an onto function?

Because all three elements of the codomain are images of elements in the domain, we see that f is onto. Note that if the codomain were $\{1, 2, 3, 4\}$, then f would not be onto.

Example

Is the function $f(x) = x^2$ from the set of integers to the set of integers onto?

Solution:

The function f is not onto because there is no integer x with $x^2 = -1$, for instance.

Example

Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3\}$ defined by $f(a) = 3$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$. Is f an onto function?

Solution: Because all three elements of the codomain are images of elements in the domain, we see that f is onto. Note that if the codomain were $\{1, 2, 3, 4\}$, then f would not be onto.

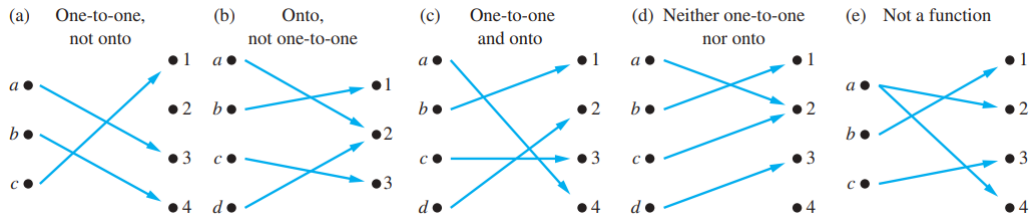


Figure: Examples of different types of correspondences.

Example

Prove that $f : \mathbb{R} - 2 \rightarrow \mathbb{R} - \{0\}$ given by $f(x) = \frac{1}{x-2}$ is onto.

$f : X \rightarrow Y$ is onto iff $\forall y \in Y \exists x \in X$ such that $f(x) = y$. **Proof:**

Let $y \in \mathbb{R} - 0$.

Then, we can write $x = \frac{1+2y}{y}$ since $y \neq 0$. Note, $x \neq 2$, if it was 2, $2 = \frac{1+2y}{y} \leftrightarrow 0 = 1$

Then,

$$\begin{aligned} f(x) &= f\left(\frac{1+2y}{y}\right) \\ &= \frac{1}{\frac{1+2y}{y} - 2} \\ &= \frac{1}{\frac{1+2y}{y} - \frac{2y}{y}} \\ &= \frac{1}{\frac{1}{y}} \\ &= y \end{aligned}$$

Scratch work done to
find x

$$\begin{aligned} f(x) &= y \\ \frac{1}{x-2} &= y \\ 1 &= xy - 2y \\ \frac{1+2y}{y} &= x \end{aligned}$$

Since $y \in \mathbb{R} - \{0\}$ was arbitrary, $\forall y \in \mathbb{R} - \{0\}$. Hence, it is onto.

Example

Prove $f : \mathbb{Z}X\mathbb{Z} \rightarrow \mathbb{Z}$ is onto if $f(m, n) = 2m - n$.

$f : X \rightarrow Y$ is onto iff $\forall y \in Y \exists x \in X$ such that $f(x) = y$.

Proof:

Suppose $y \in \mathbb{Z}$.

Then, $(0, -y) \in \mathbb{Z}X\mathbb{Z}$, and

$$\begin{aligned} f(0, -y) &= 2(0) - (-y) \\ &= y \end{aligned}$$

Rough work - can be done aside

$$f(x) = y$$

$$\begin{aligned} f(m, n) &= y \\ 2m - n &= y \end{aligned}$$

Find m and n
that when
substituted take
us to y

$$(m = 0, n = -y)$$

$$\begin{aligned} 2m - n &= 2(0) - (-y) \\ &= y \end{aligned}$$

Hence, $f(m, n) = 2m - n$ is onto.

One-to-one correspondence

The function f is a **one-to-one correspondence**, or a **bijection**, if it is both one-to-one and onto. We also say that such a function is **bijective**.

Example

Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3, 4\}$ with $f(a) = 4$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$. Is f a bijection?

Solution

The function f is one-to-one and onto. It is one-to-one because no two values in the domain are assigned the same function value. It is onto because all four elements of the codomain are images of elements in the domain. Hence, f is a bijection.

Suppose that $f : A \rightarrow B$

To show that it is injective Show that if $f(x) = f(y)$ for arbitrary $x, y \in A$, then $x = y$.

To show that f is not injective Find particular elements $x, y \in A$ such that $x \neq y$ and $f(x) = f(y)$.

To show that f is surjective Consider an arbitrary element $y \in B$ and find an element $x \in A$ such that $f(x) = y$.

To show that f is not surjective Find a particular $y \in B$ such that $f(x) \neq y$ for all $x \in A$.

Inverse Function

Definition

Let f be a one-to-one correspondence from the set A to the set B . The inverse function of f is the function that assigns to an element b belonging to B the unique element a in A such that $f(a) = b$. The inverse function of f is denoted by f^{-1} . Hence, $f^{-1}(b) = a$ when $f(a) = b$.

- A one-to-one correspondence is called **invertible** because we can define an inverse of this function.
- A function is **not invertible** if it is not a one-to-one correspondence, because the inverse of such a function does not exist.

Example

Let f be the function from $\{a, b, c\}$ to $\{1, 2, 3\}$ such that $f(a) = 2$, $f(b) = 3$, and $f(c) = 1$. Is f invertible, and if it is, what is its inverse?

Solution:

The function f is invertible because it is a one-to-one correspondence. The inverse function f^{-1} reverses the correspondence given by f , so $f^{-1}(1) = c$, $f^{-1}(2) = a$, and $f^{-1}(3) = b$.

Example

Let f be the function from $\mathbb{R}$ to $\mathbb{R}$ with $f(x) = x^2$. Is f invertible?

Solution:

Because $f(-2) = f(2) = 4$, f is not one-to-one. If an inverse function were defined, it would have to assign two elements to 4. Hence, f is not invertible.

(Note we can also show that f is not invertible because it is not onto.)

Example

Find the inverse of $f(x) = \frac{5x-3}{3x+1}$.

Sol:

$$y = \frac{5x-3}{3x+1}$$

$$3xy + y = 5x - 3$$

$$3xy - 5x = -3 - y$$

$$x(3y - 5) = -3 - y$$

$$x = \frac{-3 - y}{3y - 5}$$

$$f^{-1}(x) = \frac{-3 - x}{3x - 5}$$

Composition of Functions

Let g be a function from the set A to the set B and let f be a function from the set B to the set C . The composition of the functions f and g , denoted for all $a \in A$ by $f \circ g$, is the function from A to C defined by

$$(f \circ g)(a) = f(g(a)).$$

Example

Let g be the function from the set $\{a, b, c\}$ to itself such that $g(a) = b$, $g(b) = c$, and $g(c) = a$. Let f be the function from the set $\{a, b, c\}$ to the set $\{1, 2, 3\}$ such that $f(a) = 3$, $f(b) = 2$, and $f(c) = 1$. What is the composition of f and g , and what is the composition of g and f ?

Solution: The composition $f \circ g$ is defined by

$$(f \circ g)(a) = f(g(a)) = f(b) = 2,$$

$$(f \circ g)(b) = f(g(b)) = f(c) = 1, \text{ and}$$

$$(f \circ g)(c) = f(g(c)) = f(a) = 3.$$

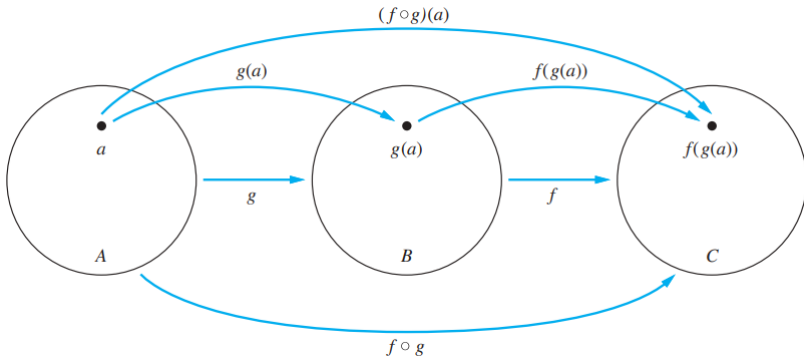


Figure: The composition of the functions f and g .

Example

Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g ? What is the composition of g and f ?

Solution:

Both the compositions $f \circ g$ and $g \circ f$ are defined. Moreover,

$$(f \circ g)(x) = f(g(x)) = f(3x + 2) = 2(3x + 2) + 3 = 6x + 7$$

and

$$(g \circ f)(x) = g(f(x)) = g(2x + 3) = 3(2x + 3) + 2 = 6x + 11.$$

Note that the commutative law does not hold for the composition of functions.

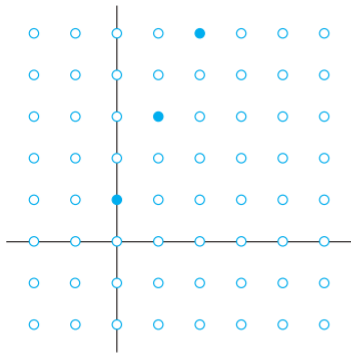
The Graphs of Functions

Definition

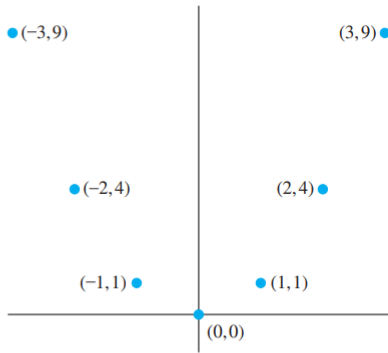
Let f be a function from the set A to the set B . The graph of the function f is the set of ordered pairs $\{(a, b) | a \in A \text{ and } f(a) = b\}$.

Example

Draw the graph of the function (1) $f(n) = 2n + 1$, and (2) $f(x) = x^2$ from the set of integers to the set of integers.



The graph of $f(n) = 2n + 1$ from $\mathbb{Z}$ to $\mathbb{Z}$.

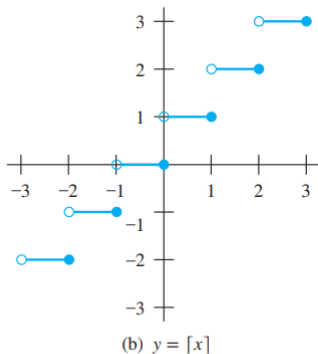
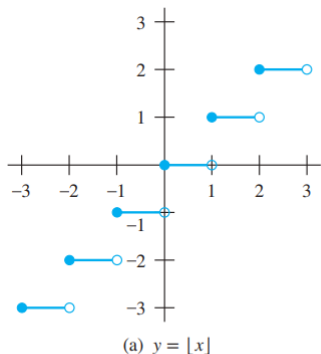


The graph of $f(x) = x^2$ from $\mathbb{Z}$ to $\mathbb{Z}$.

Floor and Ceiling Functions

The **floor function** assigns to the real number x the largest integer that is less than or equal to x . The value of the floor function at x is denoted by $\lfloor x \rfloor$.

The **ceiling function** assigns to the real number x the smallest integer that is greater than or equal to x . The value of the ceiling function at x is denoted by $\lceil x \rceil$.



Graphs of the (a) floor and (b) ceiling functions.

The End